From "Data Fragments" to "Digital Portraits"

Data Aggregation and User Privacy Erosion in the Online Advertising Ecosystem

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ABSTRACT

The core operational logic of the online advertising ecosystem is not a single dimension of "eavesdropping," but rather the aggregation of user data fragments from various channels into sellable digital portraits via Real-Time Bidding (RTB) mechanisms. Through a literature review and case study analysis, this paper systematically examines the technical architecture, industrial logic, and privacy consequences of this data aggregation process. The research reveals that 67% of Indian users report receiving relevant ads after private voice conversations, 76% experience similar phenomena after chatting on WhatsApp or social media, and 82% have granted microphone permissions to apps. However, simply attributing this phenomenon to "phone eavesdropping" is technically inaccurate. The true mechanism lies in the RTB auction triggered by every page load: a "bid request"—containing the user's advertising ID, location, device information, and interest tags—is broadcast to thousands of advertisers and data brokers. Even participants who do not win the bid can retain this data.

Keywords: Real-Time Bidding, Data Aggregation, Data Brokers, Behavioral Advertising, User Profiling, Privacy Erosion

I. INTRODUCTION

"I was just talking to a friend about wanting to buy a pair of shoes, and when I opened Instagram, I saw an ad for shoes." This is an experience almost every smartphone user has had. In India, the ubiquity of this experience is backed by survey data: 67% of respondents reported seeing relevant ads after private voice conversations—a significant rise from 53% four years ago; 76% encountered similar situations after sending messages on WhatsApp or other social platforms, with a third stating it happens "frequently."

67%	76%	82%
Report ads after private voice conversations	Report targeted ads following private chat messages	Have granted microphone access to mobile apps

Intuitively, people tend to explain this phenomenon as "the phone is listening." This intuition isn't entirely baseless—82% of Indian smartphone users have granted microphone access to apps, with 12% leaving it wide open for all apps. Google also previously settled a class-action lawsuit for $68 million over allegations that its voice assistant eavesdropped on users and shared private conversations with advertisers.

However, attributing privacy erosion strictly to "microphone eavesdropping" is technically oversimplified. If advertisers were truly transcribing the conversations of billions of users in real time, the required computational resources, bandwidth costs, and legal risks would render their business model completely unsustainable. A more accurate technical explanation is hidden inside the core infrastructure of the online advertising ecosystem: **Real-Time Bidding (RTB)**.

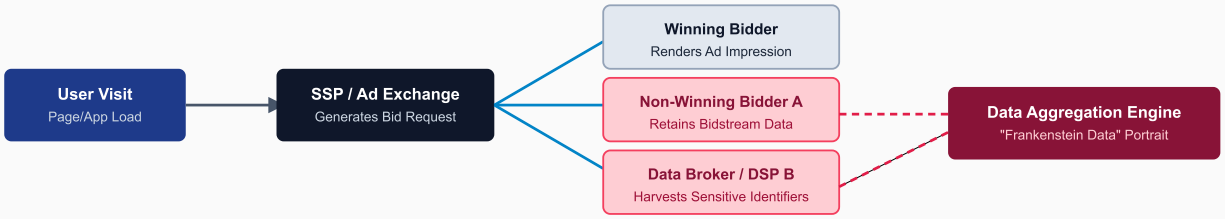
II. TECHNICAL ARCHITECTURE OF REAL-TIME BIDDING

2.1 Data Composition of a Bid Request

RTB is the underlying mechanism determining what ads you see across almost all websites and apps. Its technical workflow operates as follows: when you visit a page with ad space, the page sends a request to an ad auction operator containing details about you and the content you are viewing. The auction company packages as much information as possible into a "bid request" and broadcasts it to thousands of potential advertisers in milliseconds.

A typical bid request includes: a **unique advertising ID** (identifying your device), **location information** (GPS precision or IP-derived accuracy), **device details** (OS, browser version, device model), **interest tags** (inferred from historical behavior), **demographic info** (inferred properties like age and gender), and **User-Agent data**. Google's OpenRTB protocol further defines session parameters, including session duration, ad request depth, and requests since app launch.

FIGURE 1: REAL-TIME BIDDING (RTB) BROADCAST AND BIDSTREAM LEAKAGE ARCHITECTURE



2.2 The "Retained Even Without Winning" Loophole

A crucial yet little-known feature of the RTB architecture is that **all participating advertisers and data platforms can retain the personal data inside the bid request, regardless of whether they win the auction**. This means that even if the ad you actually see comes from a single winning bidder, your information has already been obtained by dozens or hundreds of other entities and potentially stored indefinitely.

This vulnerability serves as a primary data pipeline for data brokers. Data brokers don't need a direct relationship with website or app developers—RTB is automatically facilitated by ad companies embedded in most platforms. Data brokers simply masquerade as ad buyers participating in auctions to gain a continuous stream of data. According to analysis by the Electronic Frontier Foundation (EFF), tens of billions of RTB bid requests are broadcast daily, with thousands of ad-buying platforms receiving data from each bid.

III. THE INDUSTRIAL LOGIC OF DATA AGGREGATION: FROM FRAGMENTS TO PROFILES

3.1 The Value Chain of Data Brokers

The core business of data brokers is aggregating scattered data fragments from disparate sources into sellable user profiles. This process is termed "data enrichment"—brokers collect, analyze, and package sensitive user information to offer services like "audience segmentation" and "predictive analytics" to advertisers, employers, and law enforcement agencies.

The global data broker industry was valued at $268.73 billion in 2022 and is projected to reach nearly $365.7 billion by 2029. In India, despite rapid digitalization, data brokering remains largely unregulated, operating through third-party cookies, SDKs, and opaque partnerships.

Mechanism	Technical Operation	Privacy Impact
Aggregation & Inference	AI models compile scattered posts, search histories, and purchase logs.	Synthesizes sensitive attributes (health, politics, income).
Secondary Use	Data shared for social interaction is harvested for commercial AI.	Context is stripped (e.g., facial database from public photos).
Permanence	Data points are encoded directly into machine learning model weights.	Mathematical deletion or extraction becomes impossible.

IV. THE MOBILEWALLA CASE: AN ENFORCEMENT SAMPLE

4.1 Case Facts & Regulatory Action

In December 2024, the U.S. Federal Trade Commission (FTC) took enforcement action against data broker Mobilewalla, marking the agency's first enforcement specifically targeting the abuse of bidstream data. The FTC alleged that Mobilewalla collected data from RTB exchanges without consumer knowledge.

> **FTC v. Mobilewalla Enforcement Findings (FTC Docket No. C-4801)**
> Between 2018 and 2020, Mobilewalla harvested over **500 million unique consumer ad identifiers** paired with precise location data derived from RTB bid requests. Without anonymizing raw location coordinates, Mobilewalla profiled consumers into sensitive segments such as "pregnant women" (by tracking visits to reproductive clinics) and mapped participants of civil protests based on racial background and residency.

4.2 Enforcement Outcomes and Industry Implications

The FTC's finalized order prohibited Mobilewalla from selling sensitive location data and barred it from collecting consumer data from online ad auctions for non-bidding purposes. FTC Chair Lina Khan noted that Mobilewalla exploited loopholes in digital ad markets on an alarming scale. This case established a key legal precedent: **harvesting RTB bidstream data for non-auction profiling constitutes an unfair trade practice**.

V. USER PERCEPTION AND BEHAVIOR: THE INDIAN CONTEXT

Surveys conducted by LocalCircles (2025–2026) show that 67% of Indian respondents reported seeing relevant ads after voice conversations, and 76% after messaging on chat platforms. However, perception does not equal technical reality. Meta claims it does not use private messages for targeting, and Google settled its voice assistant litigation without admitting technical fault.

Indian users exhibit widespread permission grants: 89% share contact lists with WhatsApp, 54% with Facebook/Instagram, and 82% grant microphone access. Sharing contact lists creates a massive spillover effect, allowing platforms to profile non-users through secondary contact uploading. Furthermore, a 2026 Cashify survey revealed that 74% of consumers fear data misuse when reselling old smartphones, highlighting a systemic breakdown of trust in standard data erasure protocols.

VI. LEGAL FRAMEWORK RESPONSES & LIMITATIONS

India's Digital Personal Data Protection (DPDP) Act of 2023 (enforced via 2025 rules) established principles of purpose limitation, data minimization, and storage restriction, with penalties up to ₹2.5 billion. However, full operationalization is delayed until ~2027. More fundamentally, the DPDP Act relies on a **notice-and-consent model**. In RTB ecosystems, where data is broadcast to thousands of entities in milliseconds, traditional consent mechanisms fail entirely.

VII. CONCLUSION

The core driver of online privacy erosion is not microphone eavesdropping, but the opacity and scale of RTB data aggregation. Advertisers do not need to "listen" to voice conversations when a composite of search history, location traces, app usage, and purchase records yields an overwhelmingly accurate digital portrait. Protecting user privacy in the modern web requires moving beyond notice pop-ups toward structural accountability and strict boundaries on bidstream data processing.

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